DMS-Eval: An Empirical Benchmark of Lightweight Object Detectors for In-Cabin Driver Monitoring

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***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—driver monitoring systems, driver drowsiness detection, driver distraction detection, lightweight object detection, benchmark evaluation, computer vision, YOLO, DETR, real-time edge AI**

I. Introduction

In-cabin Driver Monitoring Systems (DMS) constitute a vital component of modern Advanced Driver Assistance Systems (ADAS) and autonomous vehicle safety suites, aiming to prevent roadway collisions through early visual detection of driver drowsiness and inattention. While high-level driver state assessment often leverages temporal recurrent or multi-frame sequence models, automotive edge deployments impose severe compute, memory, and latency constraints. Consequently, there is an urgent practical demand for lightweight, nano-scale 2D object detectors capable of executing real-time single-frame inference on embedded in-cabin hardware.

To systematically evaluate modern compact detector paradigms under strictly controlled single-frame constraints, this paper introduces **DMS-Eval**, an empirical benchmark comparing real-time nano-scale object detectors. We evaluate leading single-stage convolutional detectors (Ultralytics YOLO11n [12] and Ultralytics YOLO26n [13]) against modern end-to-end Real-Time Detection Transformers (D-FINE-N [14]).

The benchmark is developed on real-cabin RGB video from the Driver Monitoring Dataset (DMD) [15], incorporating all three original dataset session categories: *distraction*, *drowsiness*, and *gaze*. Across 81 driver-facing videos from 14 volunteer participants, frames are sampled uniformly at 1 FPS and cropped to a frozen 640×640 spatial window ($x = 272, y = 71, w = 640, h = 640$) as illustrated in Fig. 1, yielding a standardized dataset of 15,723 frames. Retaining all three session folders—particularly the *gaze* recordings and non-cue intervals—provides an abundant volume of naturalistic negative background frames (0 bounding boxes), preventing compact detectors from overtraining on positive targets and enforcing realistic false-positive suppression during routine driving (illustrated in Fig. 3a). Ground truth is constructed through 100% direct manual expert annotation in Label Studio [16] with format exports structured via `label-studio-converter` [17] targeting 4 frozen visual warning cues (exemplified in Fig. 2 and quantified in Fig. 3b):

1) `yawning`: mouth region only (wide oral opening/distension);
2) `hand_over_mouth`: full head/face enclosing the occluding hand;
3) `drinking`: driver face and beverage container together in active consumption posture;
4) `phone_use`: handheld phone and interacting hand held to ear/head in calling posture.

Under our single-frame annotation protocol, each sampled image contains at most one visual warning cue annotation (0 or 1 bounding box per image). In particular, `yawning` and `hand_over_mouth` are mutually exclusive and are never labeled twice in the same image; if a driver yawns while a hand visibly covers or occludes the mouth, the instance is uniquely labeled as `hand_over_mouth` (full head/face), ensuring clear, non-overlapping ground-truth targets.

Under the controlled-comparison principle, all models are evaluated using identical data partitions under a strict 8 / 3 / 3 subject-disjoint split (8 Train, 3 Validation, 3 Test subjects) to prevent identity leakage, utilizing identical hardware (NVIDIA RTX 4060, 8 GB VRAM), batch size 1, 220 fine-tuning epochs without early stopping, fixed random seed (13), and FP16 automatic mixed precision. Optimal checkpoints and confidence thresholds (τ) are selected strictly on the validation split prior to an isolated, single-pass test evaluation.

A. Research Question and Scope

This study is limited to **frame-level observable cues of driver drowsiness and distraction**, rather than temporal driver-state inference. Head orientation behaviors (e.g., turning away or looking toward side and rearview mirrors) are intentionally excluded from the static 2D detection ontology. Drivers routinely perform frequent mirror checks and visual scanning as part of safe driving; in isolated 1 FPS static frames without continuous temporal video context or 3D gaze tracking, there is no consistent or objective boundary to distinguish safe, brief mirror glances from dangerous, prolonged inattention. To eliminate subjective label noise from split datasets, the benchmark focuses strictly on the four unambiguous, self-contained visual warning cues.

> **RQ:** *Under a controlled, compute-constrained evaluation, how do lightweight object detectors compare in detection accuracy, inference efficiency, and robustness when identifying visual cues associated with driver distraction and drowsiness under normal and low-light/nighttime driving conditions?*

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Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable. See Table I for the primary abbreviations and acronyms used throughout the DMS-Eval benchmark.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as "3.5-inch disk drive".
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- The word "data" is plural, not singular.
- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited,

such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)

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An excellent style manual for science writers is [7].

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The class file is designed for, but not limited to, six authors. A minimum of one author is required for all conference articles. Author names should be listed starting from left to right and then moving down to the next line. This is the author sequence that will be used in future citations and by indexing services. Names should not be listed in columns nor group by affiliation. Please keep your affiliations as succinct as possible (for example, do not differentiate among departments of the same organization).

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Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

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TABLE I
LIST OF ABBREVIATIONS AND ACRONYMS

Abbreviation	Definition / Description
DMS	Driver Monitoring System
DMD	Driver Monitoring Dataset
YOLO	You Only Look Once
DETR	DEtection TRansformer
mAP	Mean Average Precision
FPS	Frames Per Second
FLOPs	Floating Point Operations
GFLOPs	Giga Floating Point Operations (10^9 FLOPs)
NMS	Non-Maximum Suppression
EAR	Eye Aspect Ratio
COCO	Common Objects in Context
IoU	Intersection over Union
AMP	Automatic Mixed Precision (FP16)
VRAM	Video Random Access Memory

b) Benchmark Status and Pending Updates: The following sections document benchmark components that are frozen alongside components awaiting empirical execution:

- **Dataset Partitions**: 14 subjects with 8/3/3 subject-disjoint split. `[TO BE UPDATED: Exact Train, Validation, and Test Subject ID allocation in splits.json]`.
- **Sampling Implementation**: Uniform 1 FPS sampling policy. `[TO BE UPDATED: Exact mapping rule from` ≈ 29.76 `FPS source video to 1 FPS extracted frames]`.
- **Software Environment**: NVIDIA RTX 4060 (8 GB VRAM). `[TO BE UPDATED: Exact pinned software versions for CUDA, PyTorch, cuDNN, Ultralytics, D-FINE commit, and THOP]`.
- **Validation Calibration**: Shared F1-maximization threshold sweep. `[TO BE UPDATED: Validation-selected optimal confidence thresholds (`$\tau_{\mathrm{YOLO11n}}, \tau_{\mathrm{D\text{-}FINE\text{-}N}}, \tau_{\mathrm{YOLO26n}}$`)]`.
- **Detection Benchmark Results**: Shared evaluator ($mAP@0.5:0.95$, $mAP@0.5$, Precision, Recall, F1). `[TO BE UPDATED: Empirical test-set detection performance metrics across candidate models]`.
- **Inference and Footprint Metrics**: FP16 median latency, throughput, THOP GFLOPs, disk size. `[TO BE UPDATED: Profiling measurements on target RTX 4060 GPU]`.

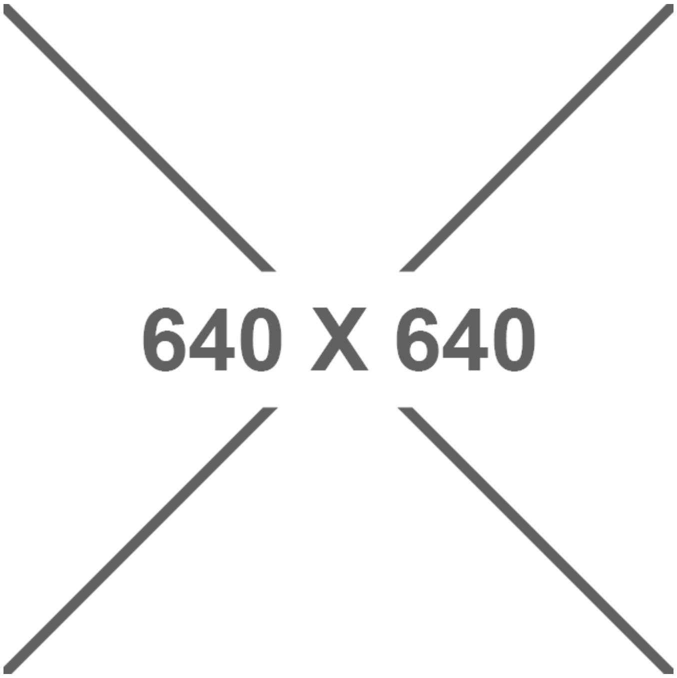

Fig. 1. In-cabin driver-facing frame extraction on DMD video with standardized 640×640 spatial window cropping ($x = 272, y = 71, w = 640, h = 640$), centering the driver's head, face, and torso for low-latency single-frame object detection.

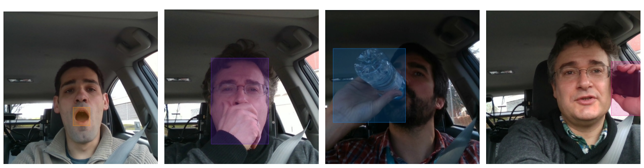

Fig. 2. Visual demonstrations of our manual annotations in Label Studio across all 4 frozen target warning cues: (a) `yawning` (orange bounding box surrounding mouth aperture only), (b) `hand_over_mouth` (purple bounding box enclosing full head, face, and occluding hand), (c) `drinking` (blue bounding box enclosing face and beverage container in active consumption posture), and (d) `phone_use` (pink bounding box enclosing hand and handheld device held at the ear in calling posture).

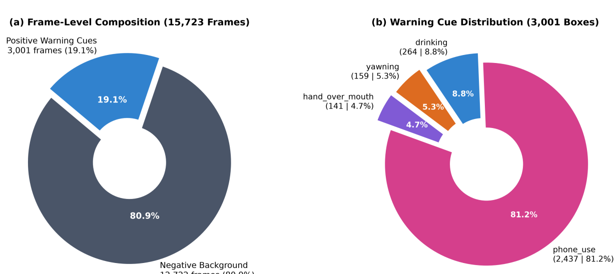

Fig. 3. DMS-Eval dataset and annotation distributions: (a) frame-level dataset composition across 15,723 frames (80.9% negative background frames vs. 19.1% positive warning cue frames); (b) proportion of bounding boxes across the 4 frozen target warning cues (3,001 total annotations: 81.2% `phone_use`, 8.8% `drinking`, 5.3% `yawning`, and 4.7% `hand_over_mouth`).

Acknowledgment

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

References

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For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [6].

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